Parametric Convex Hull Pricing for Strategic Offer Evaluation and Market Power Analysis

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***Abstract*—Convex hull pricing addresses nonconvex generation costs, but reduced lost-opportunity compensation does not by itself characterize profitable strategic offers. This paper develops a continuous-offer evaluation framework for network-constrained, multiperiod markets with complete reported lost-opportunity-cost uplift. An interval network-flow model provides the fixed pricing formulation. Common primal optima and complete endpoint duals construct a reusable selected-price map, while feasible schedule cost lines and global unit commitment bounds identify executable-schedule regions. Intersecting the partitions reduces strategic-profit evaluation to finitely many endpoint and physical-plan candidates, distinguishing attained boundary maxima from one-sided suprema. A settlement identity shows that, at a continuous selected price, pure physical switches under increasing nonnegative offers cannot raise producer profit. At an offered system-cost tie, the profit difference between executions equals their actual system-cost difference with reversed sign. Modified IEEE systems demonstrate cumulative pricing speedups of 6.74, 20.54, and 2.00 for the longer measured workloads. The principal case earns 7,811.56 dollars of additional profit, and tied executions differ by 1,387.35 dollars at identical prices. A fixed-offer network comparison shows that greater generation replacement can coexist with higher additional profit relative to each network's truthful baseline.**

***Index Terms*—Convex hull pricing, market power, parametric optimization, strategic offers, unit commitment, uplift payment.**

I. Introduction

ELECTRICITY market prices coordinate generation decisions and determine the returns to participating producers. Startup costs, minimum operating times, and ramping constraints make day-ahead unit commitment (UC) nonconvex, so uniform energy prices may leave generators with lost opportunities relative to independent operation. Uplift payments compensate these opportunities under the applicable settlement rule. Convex hull pricing (CHP) incorporates commitment nonconvexities into price formation through the convexified system cost [1], [2]. As CHP is considered for market pricing and settlement, a central market-power question is how a producer can profit by changing its offer, and which operating conditions create that opportunity.

Existing studies establish that reducing lost-opportunity costs and encouraging truthful offers are distinct market-design objectives [3], [4]. An offer change can alter nodal prices, the executable UC schedule, and the compensation calculated from reported costs. In a multiperiod network, these responses are coupled: a substitute may need to start earlier, incur additional online costs, or transfer power through a constrained line. Price sensitivity alone therefore does not determine the producer's additional profit. Repeated pointwise clearing and settlement can evaluate particular offers, but a continuous strategy range also requires locating the intervals and switching boundaries between them. The unresolved task is to evaluate that range with a consistent treatment of price selection, integer physical execution, and uplift, and then explain the economic source of the resulting gain.

This paper studies that task under network-constrained, multiperiod CHP and complete reported lost-opportunity-cost compensation. Building on an interval network-flow representation of unit operation, we follow a specified offer direction through price formation, physical clearing, and producer settlement. The resulting interval description supports reusable pricing and finite strategic-profit evaluation. It also connects profit changes to the cost of generation replacement and the transmission capacity available to that replacement. Modified IEEE systems illustrate how a producer can increase actual profit while receiving less uplift, how the timing of offer changes affects the available gain, and how transmission relaxation changes prices and feasible competition.

A. Related Work

Research on nonconvex market incentives distinguishes following an instructed dispatch from reporting actual costs. Wang *et al.* [4] analyzed these incentive properties under alternative pricing and compensation schemes. Byers and Eldridge [5] studied strategic self-commitment and self-scheduling through simulated bidder behavior in multiperiod markets. These contributions demonstrate why settlement design and bidding behavior must be assessed together. CHP-specific market-power research provides a complementary analytical foundation. Sun *et al.* [3], [6] characterized strategic gains and their relation to generator removal costs, while Wu and Sun [7] examined fixed- and variable-cost offer manipulation. Our economic question builds on these results: for a given CHP and uplift rule, which continuous offers are profitable once complete physical execution and unit self-scheduling are evaluated consistently?

The distinction between influencing prices and earning additional profit is particularly relevant across time. Sun and Wu [8] derived offer-to-price sensitivities and a temporal vulnerability assessment for CHP. Such sensitivities describe how an offer perturbation propagates through the price vector. The producer's settlement additionally depends on the executable schedule, its best self-schedule at those prices, and

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the difference between reported and actual costs. We retain this temporal price-response perspective and connect it to a profit characterization over a finite offer range. The all-day and single-hour offer directions in the numerical study expose how the admissible timing of a strategy changes its profitable opportunities.

Computing these objects requires a reliable CHP representation. Hua and Baldick [2] developed the convex-primal framework based on unit feasible-set hulls and convex cost envelopes. Extended UC formulations [9], [10] and network-flow constructions [11] attach dispatch and cost to legal operating intervals. Dantzig–Wolfe decomposition provides another exact computational route through unit-level subproblems [12]. These methods supply the convexified pricing problem used here. The network-flow foundation is retained in direct power coordinates, with the same operating and boundary conventions as physical clearing and self-scheduling. This consistency matters because changing a primal UC representation can change CHP [13].

Parametric linear programming supplies the value-function and duality properties used to reuse optimization information [14]. In the CHP literature, Jiang and Wu [15] developed a demand-to-price mapping using optimal supply sets and an approximated Minkowski sum. The present parameter enters reported cost rather than demand. Its economic output is also different: a legal price path must be combined with integer UC regions and reported-cost settlement to evaluate actual profit. Our method assembles these three objects into a continuous-offer evaluation procedure. Common primal and complete dual information establish the selected-price intervals; global UC bounds establish executable-schedule intervals; their intersection supports finite profit candidates.

B. Our Contributions

The contributions are organized around this progression from prices to executable schedules and economic outcomes.

1) *Reusable parametric CHP pricing from a network-flow model.* We express affine offer changes as objective updates in the fixed network-flow pricing LP and construct a selected-price map using common primal optima and complete optimal endpoint duals. Adaptive value bounds identify where further optimization is needed, while shared endpoint duals retain a reproducible legal price path across the range. Measured cumulative costs include map construction and online queries.
2) *Finite strategic-profit evaluation through physical UC regions.* We identify intervals with a common complete UC schedule using feasible cost lines and global mixed-integer lower bounds. Intersecting them with price segments yields regions on which actual profit is convex under complete reported uplift. This structure reduces evaluation of the continuous offer range to endpoint/physical-plan candidates. Boundary execution and one-sided branch limits are distinguished explicitly.
3) *Settlement and substitution mechanisms of profitable market power.* Combining reported self-scheduling profit with physical offer exposure characterizes nonpositive physical switching losses at continuous selected prices. At a reported system-cost tie, it also quantifies the producer's profit difference by the executions' actual system-cost difference. Cost perturbations test a predicted substitution threshold, and a network intervention compares offer-induced price and settlement changes against each network's own truthful baseline. The analysis distinguishes physical switching from profit maximization.

Section II establishes the common market and network-flow formulation. Section III develops the interval algorithms and finite-candidate result. Section IV interprets profit formation and substitution. Section V presents the computational and market evidence, followed by the conclusion.

II. Market Model and Strategic Offers

A. Physical Scheduling and Offer Direction

Let $\mathcal{G}$, $\mathcal{B}$, and $\mathcal{L}$ denote generators, buses, and lines, and let $t \in \mathcal{T} = \{1, \dots, T\}$. Unit i is connected to bus $n(i)$; g denotes the strategic unit. Its schedule is $s_i = (\boldsymbol{p}_i, \boldsymbol{u}_i) \in \mathcal{S}_i$, where $u_{i,t} \in \{0,1\}$ and $p_{i,t}$ is active power. Time intervals are one hour. The fixed unit set $\mathcal{S}_i$ imposes minimum up/down times $M_i^{\uparrow}, M_i^{\downarrow}$, inherited operating conditions, and the following output constraints (written for a generic unit g):

$$P_g^{\min} u_{g,t} \le p_{g,t} \le P_g^{\max} u_{g,t}, \tag{1a}$$

$$p_{g,t} - p_{g,t-1} \le R_g^{\uparrow} \quad (u_{g,t-1} = u_{g,t} = 1), \tag{1b}$$

$$p_{g,t-1} - p_{g,t} \le R_g^{\downarrow} \quad (u_{g,t-1} = u_{g,t} = 1), \tag{1c}$$

$$p_{g,t} \le S_g^{\uparrow} \quad (u_{g,t-1} = 0,\ u_{g,t} = 1), \tag{1d}$$

$$p_{g,t-1} \le S_g^{\downarrow} \quad (u_{g,t-1} = 1,\ u_{g,t} = 0). \tag{1e}$$

The transition at $t = 1$ uses the fixed preceding status $u_{g,0}$ and power $p_{g,0}$. Let $v_{g,t}^{\mathrm{SU}}$ and $v_{g,t}^{\mathrm{SD}}$ be binary startup and shutdown indicators. An explicit description of the commitment component is

$$u_{g,t} - u_{g,t-1} = v_{g,t}^{\mathrm{SU}} - v_{g,t}^{\mathrm{SD}}, \tag{2a}$$

$$v_{g,t}^{\mathrm{SU}} + v_{g,t}^{\mathrm{SD}} \le 1, \tag{2b}$$

$$\sum_{\tau=\max\{1,t-M_g^{\uparrow}+1\}}^{t} v_{g,\tau}^{\mathrm{SU}} \le u_{g,t}, \tag{2c}$$

$$\sum_{\tau=\max\{1,t-M_g^{\downarrow}+1\}}^{t} v_{g,\tau}^{\mathrm{SD}} \le 1 - u_{g,t}. \tag{2d}$$

The inherited residual up or down time fixes the required initial statuses. A terminal ON interval has no artificial shutdown or shutdown charge. These conventions are used unchanged in physical clearing, pricing, and self-scheduling; the chosen primal representation matters for CHP [13].

For one-hour intervals, actual operating cost is

$$C_g(s_g) = C_g^{\mathrm{SU}} n_g^{\mathrm{SU}} + C_g^{\mathrm{SD}} n_g^{\mathrm{SD}} + C_g^{\mathrm{NL}} \sum_t u_{g,t} + \sum_t \phi_g(p_{g,t} - P_g^{\min} u_{g,t}), \tag{3}$$

where $n_g^{\mathrm{SU}} = \sum_t v_{g,t}^{\mathrm{SU}}$, $n_g^{\mathrm{SD}} = \sum_t v_{g,t}^{\mathrm{SD}}$, and ϕ_g is convex piecewise linear with $\phi_g(0) = 0$. The parameter C_g^{NL} covers

operation at minimum output. For strategic unit g, define the quantity exposed to the offer increase by

$$Q_g(s_g) = \sum_t w_t(p_{g,t} - P_g^{\min} u_{g,t}), \qquad w_t \geq 0. \tag{4}$$

The reported cost is $\widehat{C}_g(s_g;\delta) = C_g(s_g) + \delta Q_g(s_g)$, with $\delta \in \mathcal{D} = [0,\bar{\delta}]$. Other units report actual cost: $\widehat{C}_i = C_i$ for $i \neq g$. The weights w_t are fixed before evaluation. The principal strategy uses $w_t = 1$ throughout the day; an hour-specific direction uses one nonzero weight. Thus Q_g is measured in MWh and δ is an absolute incremental-cost adder in USD/MWh. Physical constraints and fixed costs remain the same along this direction.

Let $H_{\ell n}$ be the PTDF and F_ℓ the line limit. With demand $D_{n,t}$, the system-feasible set $\mathcal{F}^{\mathrm{UC}}$ imposes $s_i \in \mathcal{S}_i$ and

$$\sum_i p_{i,t} = \sum_n D_{n,t}, \tag{5}$$

$$f_{\ell,t} = \sum_i H_{\ell,n(i)} p_{i,t} - \sum_n H_{\ell n} D_{n,t}, \tag{6}$$

$$-F_\ell \leq f_{\ell,t} \leq F_\ell.$$

Writing $C_{\mathrm{sys}}(s) = \sum_i C_i(s_i)$, the executable system schedule solves

$$v^{\mathrm{UC}}(\delta) = \min_{s \in \mathcal{F}^{\mathrm{UC}}} \{C_{\mathrm{sys}}(s) + \delta Q_g(s_g)\}. \tag{7}$$

The considered market is physically feasible. The independent system operator (ISO) instructs an optimal schedule $s^{\mathrm{UC}}(\delta)$, with output $\boldsymbol{p}^{\mathrm{UC}}(\delta)$. A schedule here includes the complete dispatch and commitment trajectories of all units; the cost line used below retains both.

B. CHP Through a Direct-Power Extended Formulation

CHP replaces each unit's nonconvex operating set by its joint power–commitment–cost hull, retaining the system network constraints. To make the dependence on the offer explicit, define the actual-cost epigraph

$$\mathcal{Y}_i = \{(\boldsymbol{p}_i, \boldsymbol{u}_i, \eta_i) : (\boldsymbol{p}_i, \boldsymbol{u}_i) \in \mathcal{S}_i,\ \eta_i \geq C_i(s_i)\}. \tag{8}$$

Commitment is retained in this lifted description because offer exposure (4) depends on both output and status. The convexified pricing problem is

$$v^{\mathrm{CHP}}(\delta) = \min \quad \sum_i \eta_i + \delta Q_g(\boldsymbol{p}_g, \boldsymbol{u}_g) \tag{9a}$$

$$\text{s.t.} \quad (\boldsymbol{p}_i, \boldsymbol{u}_i, \eta_i) \in \mathrm{conv}(\mathcal{Y}_i), \quad \forall i, \tag{9b}$$

$$(5),\ (6), \quad -F_\ell \leq f_{\ell,t} \leq F_\ell. \tag{9c}$$

Linear exposure can be evaluated on the convexified coordinates. Equivalently, one can convexify each reported-cost epigraph at a fixed δ. The lifted form above makes it clear why changing δ need not change the pricing feasible set.

An interval network-flow formulation represents these hulls explicitly [9]–[11]. For each unit, an alternating time-expanded directed acyclic graph connects legal ON and OFF intervals. A source-to-sink path is one commitment trajectory. Let $\mathcal{E}_i^{\mathrm{on}}$ denote ON arcs and B_i the signed node–arc incidence matrix. With one unit of supply at the source and one unit of demand at the sink, the arc weights satisfy

$$B_i z_i = b_i^{\mathrm{flow}}, \qquad z_i \geq 0. \tag{10}$$

Minimum operating times and inherited conditions determine the allowed arcs. Fractional flow decomposes into commitment paths: an ON arc can be shared by several full trajectories, avoiding explicit enumeration of every complete trajectory.

For an ON arc $e = [a,b]$, introduce scaled power $p_{g,e,t}$ directly. Its perspective dispatch constraints include

$$P_g^{\min} z_{g,e} \leq p_{g,e,t} \leq P_g^{\max} z_{g,e}, \quad a \leq t \leq b, \tag{11a}$$

$$p_{g,e,t} - p_{g,e,t-1} \leq R_g^{\uparrow} z_{g,e}, \quad a < t \leq b, \tag{11b}$$

$$p_{g,e,t-1} - p_{g,e,t} \leq R_g^{\downarrow} z_{g,e}, \quad a < t \leq b. \tag{11c}$$

A new startup gives $p_{g,e,a} \leq S_g^{\uparrow} z_{g,e}$; a following shutdown gives $p_{g,e,b} \leq S_g^{\downarrow} z_{g,e}$. For an inherited ON arc beginning at $t = 1$, the initial ramp inequalities instead use $p_{g,0} z_{g,e}$. These are direct scaled powers, not changes in power. If $\phi_g(q) = \max_k\{a_{gk}q + b_{gk}\}$, interval cost variables satisfy

$$\xi_{g,e,t} \geq a_{gk}(p_{g,e,t} - P_g^{\min} z_{g,e}) + b_{gk} z_{g,e}, \quad \forall k. \tag{12}$$

Let $\mathcal{E}_i$ denote all ON and OFF arcs, and let $\kappa_{i,e}$ collect the fixed costs assigned to arc e, including online duration and actual state-transition charges. Output, status, and cost use the same arc weights:

$$p_{i,t} = \sum_{e \in \mathcal{E}_i^{\mathrm{on}}: t \in e} p_{i,e,t}, \tag{13a}$$

$$u_{i,t} = \sum_{e \in \mathcal{E}_i^{\mathrm{on}}: t \in e} z_{i,e}, \tag{13b}$$

$$\eta_i \geq \sum_{e \in \mathcal{E}_i} \kappa_{i,e} z_{i,e} + \sum_{e \in \mathcal{E}_i^{\mathrm{on}}} \sum_{t \in e} \xi_{i,e,t}. \tag{13c}$$

Inherited immediate-shutdown charges, when applicable, are assigned to the corresponding initial OFF arc. They are included with the fixed arc costs in the same aggregation. At zero arc weight, capacity forces its output to zero. At positive weight, division by that weight recovers an ordinary feasible ON-interval trajectory and its cost. Flow decomposition then mixes those trajectories under common power and cost weights. This supplies the joint hull for the stated interval-local model; its construction is the computational foundation inherited from the network-flow approach.

For fixed physical parameters and an affine offer direction, the pricing problem has the form

$$v^{\mathrm{CHP}}(\delta) = \min_{x \in \mathcal{P}} (c^0 + \delta d)^\top x, \tag{14}$$

$$\mathcal{P} = \{x : Ax = b,\ Gx \leq h,\ \underline{x} \leq x \leq \overline{x}\}.$$

Here x collects arc-flow, direct-power, and cost variables. The offer direction has the explicit action

$$d^\top x = \sum_t w_t \sum_{e \in \mathcal{E}_g^{\mathrm{on}}: t \in e} (p_{g,e,t} - P_g^{\min} z_{g,e}). \tag{15}$$

Consequently, $\mathcal{P}$ and the network matrices are fixed, and each new offer updates only objective coefficients. A change in demand, physical line limits, unit data, or inherited operating

conditions defines a different instance whose map must be rebuilt or revalidated. The convexified pricing output $\boldsymbol{p}^{\mathrm{CHP}}$ supports prices and can be fractional in commitment; the executable output remains $\boldsymbol{p}^{\mathrm{UC}}$ from (7). The next subsection defines the independent unit response needed to turn these prices into settlement.

Let λ_t^E be the multiplier on system balance, with nonnegative multipliers $\mu_{\ell,t}^+$ and $\mu_{\ell,t}^-$ on the upper and lower line limits. Nodal CHP is

$$\lambda_{n,t}^{\mathrm{CHP}} = \lambda_t^E + \lambda_{n,t}^C, \qquad \lambda_{n,t}^C = -\sum_{\ell} H_{\ell n}(\mu_{\ell,t}^+ - \mu_{\ell,t}^-). \quad (16)$$

The system energy component is the balance multiplier of the constrained problem. Both components depend on the PTDF reference; nodal prices and settlement are invariant under a consistent reference change.

C. Self-Scheduling, Uplift, and Actual Profit

Given the selected nodal price vector $\boldsymbol{\lambda}_g = (\lambda_{n(g),t}^{\mathrm{CHP}})_{t\in\mathcal{T}}$, a generator's best reported self-scheduling profit is

$$\widehat{\Pi}_g^{\mathrm{SS}}(\boldsymbol{\lambda}_g, \delta) = \max_{s_g \in \mathcal{S}_g} \{\boldsymbol{\lambda}_g^\top \boldsymbol{p}_g - \widehat{C}_g(s_g; \delta)\}. \quad (17)$$

Its optimizer s_g^{SS} satisfies unit operating constraints; the collection of unit best responses need not balance system demand or satisfy network limits. This is the third dispatch object, distinct from physical clearing and convexified pricing output.

Actual energy revenue, uplift payment, and actual profit are

$$R_g(\delta) = \boldsymbol{\lambda}_g^{\mathrm{CHP}}(\delta)^\top \boldsymbol{p}_g^{\mathrm{UC}}(\delta), \quad (18)$$

$$U_g(\delta) = \widehat{\Pi}_g^{\mathrm{SS}} - [R_g(\delta) - \widehat{C}_g(s_g^{\mathrm{UC}}; \delta)], \quad (19)$$

$$\Pi_g(\delta) = R_g(\delta) + U_g(\delta) - C_g(s_g^{\mathrm{UC}}(\delta)). \quad (20)$$

Throughout this paper, uplift means the complete reported lost-opportunity-cost payment in (19). Since s_g^{UC} is feasible for (17), $U_g \geq 0$. Reported cost determines compensation, whereas actual operating cost determines actual profit.

For fixed other offers, the additional-profit objective is

$$\mathcal{M}_g = \max_{\delta\in[0,\bar{\delta}]} \Delta\Pi_g(\delta), \qquad \Delta\Pi_g(\delta) = \Pi_g(\delta) - \Pi_g(0). \quad (21)$$

The maximum is evaluated for a specified price and physical execution rule. We construct a legal price selection by complete-dual interpolation and use a common optimal schedule on each open physical region. At a boundary, both adjacent schedules are evaluated. Taking the higher-profit adjacent schedule defines the upper boundary candidate used in the main comparison. This is a potential-profit evaluation rule, not a producer instruction to the ISO. Section III distinguishes its attained maximum from the one-sided supremum when a different adjacent schedule is executed at the boundary.

III. Parametric Pricing and Finite Offer Evaluation

The computational task is to evaluate actual profit over $\mathcal{D}$, following both a legal CHP price selection and executable physical clearing. The two optimization problems share the same scalar offer but have different feasible sets: the pricing problem is an LP on the unit hulls, whereas physical clearing retains integer commitment. We first construct price intervals, then find common physical schedules, and finally combine them for settlement. The finite-candidate result is the endpoint of this construction.

A. Complete Duals and Affine Selected Prices

The value of (14) is concave and piecewise affine in δ, a standard property of objective-parametric linear programs [14]. Its price vector is a projection of an optimal dual solution and need not be unique. The object constructed here is one legal selected-price path, supported by the full primal–dual information.

Write a complete LP dual as $y = (\pi, \nu, \zeta^L, \zeta^U)$. Its feasible set $\mathcal{F}^D(\delta)$ is defined by

$$A^\top\pi + G^\top\nu + \zeta^L + \zeta^U = c^0 + \delta d, \quad (22a)$$

$$\nu \leq 0, \qquad \zeta^L \geq 0, \qquad \zeta^U \leq 0, \quad (22b)$$

where π is free. Multipliers for absent infinite bounds are fixed to zero. The dual objective is

$$L(y) = b^\top\pi + h^\top\nu + \underline{x}^\top\zeta^L + \overline{x}^\top\zeta^U, \quad (23)$$

with the bound terms summed over finite bounds only. Since all right-hand sides are fixed, L is independent of δ. The positive line multipliers in (16) equal the negatives of the corresponding entries of ν. Retaining bound multipliers as well as row multipliers makes stationarity and strong duality checkable for the actual network-flow LP.

Proposition 1 (Affine selected prices)**.** *Suppose one feasible $\bar{x}$ is optimal at $a < b$, and complete dual optima $y^a \in \mathcal{F}^D(a)$ and $y^b \in \mathcal{F}^D(b)$ attain its endpoint costs. For $\delta = (1-\theta)a + \theta b$, $0 \leq \theta \leq 1$, the vector $y(\delta) = (1-\theta)y^a + \theta y^b$ is dual optimal. Its nodal-price projection is affine on $[a, b]$.*

Proof. Interpolating (22) gives stationarity for $c^0 + \delta d$ and preserves every multiplier sign. Furthermore,

$$\begin{aligned} L(y(\delta)) &= (1-\theta)L(y^a) + \theta L(y^b) \\ &= (c^0 + \delta d)^\top \bar{x}. \end{aligned}$$

Primal and dual feasibility with equal objectives establish optimality. Applying the linear price projection in (16) gives the stated affine path. □

This result uses the common primal optimum rather than a particular simplex basis. Degeneracy may permit several optimal bases and several legal prices; the complete endpoint duals select the path used for economic evaluation. Adjacent accepted intervals share their endpoint dual. Their price interpolation is therefore continuous at that endpoint even if the common primal solution changes.

B. Adaptive Construction of Price Intervals

At a solved offer δ_i, retain a feasible primal x^i, a complete optimal dual y^i, and the coefficients of its affine feasible-cost line

$$\ell_i^P(\delta) = A_i^P + \delta B_i^P, \quad A_i^P = (c^0)^\top x^i, \quad B_i^P = d^\top x^i. \quad (24)$$

For a bracket $[a, b]$, endpoint lines and dual values give

$$U^P_{ab}(\delta) = \min\{\ell^P_a(\delta), \ell^P_b(\delta)\}, \tag{25}$$

$$L^P_{ab}(\delta) = \frac{b-\delta}{b-a} L(y^a) + \frac{\delta-a}{b-a} L(y^b). \tag{26}$$

Every feasible primal line upper-bounds v^{CHP}, while the convex combination of endpoint duals is feasible at the interpolated objective and supplies a lower bound. Thus $L^P_{ab} \le v^{\mathrm{CHP}} \le U^P_{ab}$.

The gap $U^P_{ab} - L^P_{ab}$ identifies where another solve can improve the interval description. With two endpoint lines, an interior maximizer occurs at their crossing when that crossing lies strictly inside the bracket; otherwise the procedure uses its midpoint. Among unresolved brackets, the largest remaining gap is refined first. Each new solve updates the fixed LP's objective, obtains a new complete primal–dual pair, and splits the bracket.

A small value gap alone is not the price certificate. For the pool $\mathcal{W}^P$ of saved feasible primals, the procedure also tests the common-primal discrepancy

$$\gamma^P_{ab} = \min_{x \in \mathcal{W}^P} \max_{q \in \{a,b\}} \left| (c^0 + qd)^\top x - L(y^q) \right|. \tag{27}$$

In exact arithmetic, $\gamma^P_{ab} = 0$ supplies the common optimum of Proposition 1. Numerically, the endpoint primal feasibility, dual signs, stationarity, complementarity, and duality gap are checked, followed by the specified objective-gap tolerance. An interval opens for price queries only after these checks pass. A partial map retains its unresolved brackets if the solve budget is reached.

The selected-price intervals and value intervals can then be compressed separately. A union of value intervals is merged when a single feasible primal remains optimal at its outer endpoints. A selected-price breakpoint can be removed when the nodal prices remain collinear under the original selection. These operations preserve different objects, explaining why price and value segment counts can differ.

C. Constructing Common Physical UC Regions

Physical clearing requires its own interval search. A feasible complete system schedule s^j defines

$$\ell^U_j(\delta) = A^U_j + \delta B^U_j, \quad A^U_j = C_{\mathrm{sys}}(s^j), \quad B^U_j = Q_g(s^j_g). \tag{28}$$

It is feasible for every δ because the offer changes cost only. Consequently, v^{UC} is the lower envelope of these schedule lines over all feasible schedules. It is concave even though the schedules contain integer commitments.

Lemma 1 (Common physical schedule)**.** *If a system-feasible schedule is optimal at both a and b, it is optimal throughout $[a, b]$.*

Proof. Its affine cost line upper-bounds v^{UC}. Concavity lower-bounds v^{UC} by the chord of its endpoint values. When the schedule is optimal at both endpoints, its line equals that chord. □

To find such schedules, solve the physical UC at the domain endpoints, retaining complete feasible trajectories and valid global mixed-integer lower bounds $\underline{v}^U(a)$ and $\underline{v}^U(b)$. For the current schedule pool $\mathcal{W}^U$, choose the best common witness using

$$\gamma^U_{ab} = \min_{s^j \in \mathcal{W}^U} \max_{q \in \{a,b\}} \left| \ell^U_j(q) - \underline{v}^U(q) \right|. \tag{29}$$

A zero discrepancy establishes the premise of Lemma 1. For a numerical tolerance ε_U, it measures how closely one complete feasible schedule meets the global endpoint bounds. If schedule j passes, the lower chord of these bounds implies, for $\delta = (1-\theta)a + \theta b$,

$$\begin{aligned} 0 \le \ell^U_j(\delta) - v^{\mathrm{UC}}(\delta) &\le (1-\theta)[\ell^U_j(a) - \underline{v}^U(a)] \\ &\quad + \theta[\ell^U_j(b) - \underline{v}^U(b)] \le \varepsilon_U. \end{aligned} \tag{30}$$

Thus the same complete feasible schedule has a uniformly bounded objective discrepancy across the bracket. This is a cost certificate; it does not turn a small objective gap into a dispatch or profit error bound.

If no common schedule passes, let s^a, s^b be the schedules returned at the bracket endpoints. Their candidate crossing is

$$q_{ab} = \frac{A^U_b - A^U_a}{B^U_a - B^U_b}, \tag{31}$$

when the denominator is nonzero. Solve the global UC there if $q_{ab} \in (a, b)$; use the midpoint otherwise. A new competing schedule may enter and become a witness on one of the subintervals. The crossing proposes a useful next offer, while global UC lower bounds determine whether either candidate belongs to the optimal envelope. A better third schedule can displace both incumbents and is retained when the new global solve identifies it.

The implementation refines the bracket with the largest common-schedule discrepancy, repeats the endpoint tests on its subdivisions, and merges adjacent accepted intervals only when one saved schedule covers their union. It therefore identifies executable-schedule regions adaptively rather than inferring them from a dense predetermined offer grid. A full-domain assessment proceeds only after the accepted regions cover $\mathcal{D}$ without gaps.

D. Joint Regions and Finite Profit Candidates

Intersect the selected-price partition and common physical partition. Each joint interval $I_j = [a_j, b_j]$ has an affine selected price $\boldsymbol{\lambda}_g(\delta) = \alpha^j_g + \delta\beta^j_g$ and a common complete system schedule $s^{\mathrm{UC},j}$. The interval profit Π^j_g evaluates that physical schedule throughout the interval, including its endpoint extension.

Proposition 2 (Finite profit evaluation)**.** *For a fixed unit feasible set and actual cost, complete reported uplift, an affine selected price on I_j, and a common physical schedule, Π^j_g is convex and*

$$\max_{\delta \in [a_j, b_j]} \Pi^j_g(\delta) = \max\{\Pi^j_g(a_j), \Pi^j_g(b_j)\}. \tag{32}$$

A finite covering joint partition therefore reduces the maximum of these branch profits to its endpoint/physical-plan candidates.

Proof. Substituting (19) into (20) gives

$$\Pi_g^j(\delta) = \max_{s_g \in \mathcal{S}_g} \{a_g^j(s_g) + \delta b_g^j(s_g)\} + \delta Q_g(s_g^{\mathrm{UC},j}), \quad (33)$$

where $a_g^j(s_g) = (\alpha_g^j)^\top \boldsymbol{p}_g - C_g(s_g)$ and $b_g^j(s_g) = (\beta_g^j)^\top \boldsymbol{p}_g - Q_g(s_g)$. Every fixed self-schedule contributes an affine function of δ. Their supremum is convex; the last term is affine. A convex function on a compact interval is bounded above by the larger endpoint value, which proves (32). Applying it to every covering interval gives the finite reduction. □

Let $\mathcal{C} = \{(a_j, s^{\mathrm{UC},j}), (b_j, s^{\mathrm{UC},j}) : j\}$, retaining different physical plans at a shared offer. The candidate value is

$$\mathcal{M}_g^{\mathrm{cand}} = \max_{(\delta,s)\in\mathcal{C}} \{\Pi_g(\delta; s) - \Pi_g(0)\}. \quad (34)$$

Unit self-scheduling is solved at these candidates to compute reported uplift and actual profit. One need not construct all optimal self-scheduling regions to find this value. Common self-schedules are needed only if a complete profit/uplift curve or a local channel decomposition is also desired.

The boundary rule determines whether the candidate value is attained or approached. The upper adjacent-plan rule takes the more profitable adjacent physical plan at a shared boundary, so $\mathcal{M}_g = \mathcal{M}_g^{\mathrm{cand}}$ under this rule. Write δ^Π for a maximizing offer. For another execution rule ρ that follows the same common schedules in open intervals and chooses an adjacent plan at each boundary, continuity of each branch gives

$$\sup_{\delta\in\mathcal{D}} [\Pi_g^\rho(\delta) - \Pi_g(0)] = \mathcal{M}_g^{\mathrm{cand}}. \quad (35)$$

A maximizing boundary value need not be attained under that rule, but can be approached from its corresponding interval. Here branch continuity follows from the bounded unit output set and continuous cost in the self-scheduling problem. Every interior or adjacent-boundary execution is bounded by (34); approaching a maximizing endpoint from its open interval gives the reverse inequality in (35). At $\delta = 0$, the common selected price and complete reported uplift make actual profit independent of physical tie selection. Equation (35) concerns these specified branch executions, not a choice among every system-optimal schedule at every offer. The numerical study reports both the upper boundary candidate and a directly settled nearby interior offer.

E. Implementation and Computational Scope

The pricing model retains its sparse network-flow matrices, variable indices, and offer direction. Only the objective is updated at a new δ; all primal and dual solution arrays are then extracted anew. Each physical UC solve supplies an incumbent complete schedule and a global bound. Interval certificates, physical plans, and selected endpoint duals are saved so interrupted construction can resume without changing the declared economic objects. Failure to complete coverage produces a partial result, as indicated in Algorithm 1.

For a completed map with K selected-price segments, an online query uses $O(\log K)$ comparisons to locate the segment and affine arithmetic to return prices. Offline work depends on the discovered structure and LP/MIP solves; neither the full region count nor the integer-solve effort is claimed to be polynomial in the horizon. The measured pricing comparison therefore includes map construction, postprocessing, loading, and query costs. It is separate from the physical-region and self-scheduling computation used for profit evaluation.

Algorithm 1 Parametric CHP and Profit Evaluation

Require: Fixed market data, target g, weights w, domain $\mathcal{D}$
Solve endpoint CHP LPs; retain full primal–dual pairs
while unresolved price brackets remain and budget permits **do**
  Test a common primal using (27)
  Accept by Proposition 1, or solve at the value-gap probe
end while
if accepted price intervals do not cover $\mathcal{D}$ **then**
  return partial mapping and unresolved brackets
end if
Solve endpoint physical UCs; retain schedules and global bounds
while unresolved physical brackets remain and budget permits **do**
  Select the bracket with largest γ_{ab}^U
  if a common schedule passes (29) **then**
    Accept the bracket with this complete schedule
  else
    Form cost-line crossing (31); use midpoint if invalid
    Solve global UC at the probe and subdivide
  end if
end while
if accepted physical intervals do not cover $\mathcal{D}$ **then**
  return checkpoint and unresolved physical intervals
end if
Merge only where a common witness covers the union
Intersect price and physical partitions to form $\mathcal{C}$
for each distinct endpoint/physical-plan pair in $\mathcal{C}$ **do**
  Solve reported self-scheduling; calculate R_g, U_g, C_g, Π_g
end for
return $\mathcal{M}_g^{\mathrm{cand}}$, maximizing candidates, and side limits

The resulting method uses the network-flow CHP solver in a specific way: continuous offers are organized around shared optimality information, while executable UC regions supply the dispatch selection that settlement requires. The next section interprets the finite candidates economically, explaining how price effects accumulate within a region and how generation replacement changes profit at its boundaries.

IV. Profit Formation and Physical Substitution

Section III identifies the prices and physical plans associated with profitable candidates. Their economic interpretation requires separating the value of the announced prices from the cost discrepancy created by the offer. This section derives that separation and connects physical switching to the expense of competing generation.

A. Profit Identity and Local Channels

The algebra used in Proposition 2 also gives an economic characterization applicable to any feasible physical execution:

$$\boxed{\Pi_g(\delta) = \widehat{\Pi}_g^{\mathrm{SS}}(\boldsymbol{\lambda}_g(\delta), \delta) + \delta Q_g(s_g^{\mathrm{UC}}(\delta)).} \tag{36}$$

The first term is the highest reported operating profit at the announced prices. The second restores the gap between reported and actual costs on the instructed schedule. At truthful bidding, actual profit equals the best self-scheduling profit. At positive adders, physical selection enters through the offer-exposed quantity.

For two optimal physical schedules A and B at the same offer and selected price, (36) implies

$$\Pi_g^A - \Pi_g^B = \delta\,[Q_g(s_g^A) - Q_g(s_g^B)]. \tag{37}$$

The common reported self-scheduling value cancels between the two executions. Let C_A, C_B be their actual total system costs. With all other units reporting actual costs, the offered system-cost tie gives $C_A + \delta Q_A = C_B + \delta Q_B$, and hence

$$\boxed{\Pi_g^A - \Pi_g^B = C_B - C_A.} \tag{38}$$

Thus, identical offers, selected prices, and optimal reported system costs can conceal different actual producer profits. The execution with lower actual system cost gives this producer the higher profit. At truthful bidding both differences vanish. This conclusion uses full reported uplift; its magnitude is determined by the complete physical schedules.

Within a differentiable joint region, an optimal self-schedule gives the envelope relation

$$\begin{aligned}\frac{d\Pi_g}{d\delta} &= \underbrace{(\boldsymbol{p}_g^{\mathrm{SS}})^\top \frac{d\boldsymbol{\lambda}_g^{\mathrm{CHP}}}{d\delta}}_{\text{price effect}} + \underbrace{Q_g^{\mathrm{UC}} - Q_g^{\mathrm{SS}}}_{\text{offer-exposure effect}} \\ &= (\boldsymbol{p}_g^{\mathrm{SS}})^\top \frac{d\boldsymbol{\lambda}^E}{d\delta} + (\boldsymbol{p}_g^{\mathrm{SS}})^\top \frac{d\boldsymbol{\lambda}_g^C}{d\delta} + Q_g^{\mathrm{UC}} - Q_g^{\mathrm{SS}}.\end{aligned} \tag{39}$$

At self-scheduling switches, the relation applies on differentiable pieces, which are sufficient for integration. If the selected price is continuous and physical execution changes at b, its actual-profit jump is

$$J_g(b) = b[Q_g^{\mathrm{UC},+}(b) - Q_g^{\mathrm{UC},-}(b)]. \tag{40}$$

The sign of this jump follows from optimal physical clearing, independently of which constraints produce the replacement.

Corollary 1 (Offer exposure and physical switching)**.** *Fix the physical feasible set and all other offers. For $\delta_1 < \delta_2$, any globally optimal executions satisfy*

$$Q_g^{\mathrm{UC}}(\delta_2) \le Q_g^{\mathrm{UC}}(\delta_1). \tag{41}$$

At a nonnegative switching offer b with a continuous selected CHP price, complete reported uplift therefore gives $J_g(b) \le 0$. If other offers equal actual costs, adjacent complete schedules additionally satisfy

$$J_g(b) = -(C_+ - C_-) \le 0. \tag{42}$$

Proof. Write the fixed objective intercept as $C_0(s)$. Optimality at the two offers gives $C_0(s_1) + \delta_1 Q_1 \le C_0(s_2) + \delta_1 Q_2$ and $C_0(s_2) + \delta_2 Q_2 \le C_0(s_1) + \delta_2 Q_1$. Adding yields $(\delta_2 - \delta_1)(Q_2 - Q_1) \le 0$. Take adjacent limits and apply (40). At the boundary, equality of the two offered cost lines gives $C_+ - C_- = b(Q_- - Q_+)$ when C_0 is actual system cost. □

The optimality argument is a standard property of affine objective perturbations [14]. Its settlement implication identifies what replacement can do: a pure physical switch cannot create a positive profit jump under these assumptions, although price responses between switches may increase profit. A discontinuous price selection requires the additional self-scheduling value change described below.

Along a path from truthful bidding to offer δ_f, the full change is

$$\begin{aligned}\Delta\Pi_g(\delta_f) = &\int_0^{\delta_f} (\boldsymbol{p}_g^{\mathrm{SS}})^\top \frac{d\boldsymbol{\lambda}_g^{\mathrm{CHP}}}{d\delta}\, d\delta \\ &+ \int_0^{\delta_f} (Q_g^{\mathrm{UC}} - Q_g^{\mathrm{SS}})\, d\delta + \sum_{b \in \mathcal{K}(\delta_f)} J_g(b).\end{aligned} \tag{43}$$

Here $\mathcal{K}(\delta_f)$ includes the execution changes traversed by that path, including a terminal change only if the selected endpoint execution requires it. The upper candidate at a switch can therefore be evaluated by its incoming branch without adding the outgoing branch's drop. If an alternative price rule has a jump, its additional term is the change of $\widehat{\Pi}_g^{\mathrm{SS}}$ between the two prices at the same offer, rather than a derivative approximation. The shared-endpoint price rule used here is continuous.

Equation (39) weights prices by the best self-schedule, whereas energy revenue in (18) uses actual cleared output. This explains why a positive price-related profit effect and the change in energy revenue have different magnitudes. It also connects temporal price vulnerability to a settlement-based profit measure without equating the two.

B. Substitution Costs and Physical Switching Thresholds

Let δ^{UC} denote a threshold at which the optimal physical schedule changes, and retain δ^{Π} for a profit-maximizing offer under the specified execution rule. Consider competing complete schedules A, B with actual system costs C_A, C_B and target exposures $Q_A > Q_B$. Their reported cost lines meet at

$$\delta^{\mathrm{UC}} = \frac{C_B - C_A}{Q_A - Q_B}. \tag{44}$$

The numerator measures the additional actual system cost of the alternative; the denominator measures the offer-exposed generation it replaces. Below this crossing, the target's larger exposure is offset by the lower actual system cost of schedule A. Above it, the extra reported cost favors schedule B. Interpreting (44) as an actual clearing threshold requires that no third schedule dominate near the crossing, which is checked using the global UC bounds in Section III.

The threshold δ^{UC} follows from physical cost competition; δ^{Π} additionally depends on CHP and uplift. They need not coincide. When price-related gains accumulate on a physical region but replacement causes a downward profit jump at its boundary, the incoming side of δ^{UC} becomes an economically relevant candidate for δ^{Π}. The finite evaluation compares this

candidate with every other joint-region endpoint, establishing whether it is the most profitable one.

If a specified cost perturbation changes schedule j by $\epsilon\chi_j$, the candidate threshold becomes

$$\delta^{\mathrm{UC}}(\epsilon) = \delta^{\mathrm{UC}}(0) + \epsilon\frac{\chi_B - \chi_A}{Q_A - Q_B}. \tag{45}$$

For an online-cost perturbation, χ_j is the substitute's online duration. Equal startup counts instead predict no threshold movement under a startup-cost perturbation. These directions distinguish additional online operation from the cost of initiating that operation.

C. Transmission Prices and Feasible Competition

Network constraints can affect both terms involved in strategic opportunity. They shape the optimal dual prices in (16) and restrict the alternative schedules available in (28). Energy/congestion accounting identifies price components under a fixed reference. A controlled change to a line limit additionally reveals changes in feasible competition. These two analyses are used together.

Physical supply dependence, price influence, and additional profit are distinct objects. Minimum required output measures feasible generation; Q_g measures the specific incremental quantity exposed to the offer. A unit required to operate near minimum output can have substantial physical dependence but little mandatory incremental exposure. Neither measure alone determines the price or self-scheduling terms in (36).

For a network scenario S, define $\Delta_\delta\lambda^S = \lambda^S(\delta) - \lambda^S(0)$ and $\Delta\Pi_g^S = \Pi_g^S(\delta) - \Pi_g^S(0)$. Comparing relaxed ($r$) and original ($o$) networks uses

$$\mathcal{I}^{\lambda}_{n,t}(\delta) = \Delta_\delta\lambda^r_{n,t} - \Delta_\delta\lambda^o_{n,t}, \tag{46}$$

$$\begin{aligned}\mathcal{I}^{\Pi}_{g}(\delta) &= \Delta\Pi^r_g - \Delta\Pi^o_g \\ &= (\Delta R^r_g - \Delta R^o_g) + (\Delta U^r_g - \Delta U^o_g) \\ &\quad - (\Delta C^r_g - \Delta C^o_g).\end{aligned} \tag{47}$$

These deterministic contrasts measure how the same offer changes prices and settlement relative to each network's baseline. The terms are complete accounting contributions; they do not assign independent causal shares to prices and physical constraints.

For settlement consistency, define

$$\Delta^{\mathrm{TX}} = \sum_{\ell,t}[\mu^+_{\ell,t}(F_\ell - f^{\mathrm{UC}}_{\ell,t}) + \mu^-_{\ell,t}(F_\ell + f^{\mathrm{UC}}_{\ell,t})]. \tag{48}$$

To see the settlement relation, let $D^\lambda = \sum_{n,t}\lambda^{\mathrm{CHP}}_{n,t}D_{n,t}$. Strong duality and unit self-scheduling give

$$v^{\mathrm{CHP}} = D^\lambda - \sum_{\ell,t}(\mu^+_{\ell,t} + \mu^-_{\ell,t})F_\ell - \sum_i \widehat{\Pi}^{\mathrm{SS}}_i. \tag{49}$$

Equation (16) and physical balance also give $\sum_i R_i - D^\lambda = -\sum_{\ell,t}(\mu^+_{\ell,t} - \mu^-_{\ell,t})f^{\mathrm{UC}}_{\ell,t}$. Substitution into (19) yields

$$\sum_i U_i + \Delta^{\mathrm{TX}} = v^{\mathrm{UC}}(\delta) - v^{\mathrm{CHP}}(\delta). \tag{50}$$

The values on the right use reported costs. The nonnegative transmission adjustment evaluates pricing multipliers against physical scheduled flows; it is distinct from a nodal congestion-price component or an individual generator's uplift. In particular, a binding physical line need not have a nonzero CHP multiplier.

TABLE I
PRICE MAPPING AND MEASURED QUERY PERFORMANCE

Instance	Price segments	Queries	Direct (s)	Param. (s)	Ratio
30-G1	83	1000	286.46	42.52	6.74
30-G2	8	1000	119.28	5.81	20.54
118-G11	103	500	623.41	311.60	2.00
118-G11	103	200	279.35	311.60	0.90

Ratio is direct/parametric cumulative time. The 118-bus rows are independent workloads on the same archived mapping. G1 has 17 common physical regions, 99 joint intervals, and 116 endpoint/physical-plan candidates at 100 distinct offers.

V. NUMERICAL RESULTS

The studies use modified IEEE 30- and 118-bus systems with a 24-hour horizon, three incremental cost segments, and PTDF transmission constraints. The systems have six and 54 generators, respectively. All units are initially offline at zero output with minimum down times satisfied. Every optimization and settlement uses these same inherited conditions. The reference incremental slopes for the principal G1 case and the 118-bus G11 case are 41.982 and 36.33 USD/MWh. The 0–50% normalized ranges therefore correspond to absolute adders $[0, 20.991]$ and $[0, 18.165]$ USD/MWh; percentages describe the offer parameter.

The archived timing workloads use COPT 8.0.6 and Python 3.13.5 on the same workstation. Persistent reoptimization includes setup, objective updates, optimization, and price extraction. Parametric totals include construction, postprocessing, loading, and lookup. Queries are fixed shuffled sequences. Physical UC and unit self-scheduling are evaluated separately for settlement; their cost is not included in pricing speedup ratios. Pricing intervals use complete-dual checks and a 10^{-6}-dollar common-primal acceptance tolerance. Physical regions use global UC bounds with a 10^{-5}-dollar cost tolerance. These are objective-gap acceptance tolerances, not bounds on dispatch or profit error. The archived checks and source fingerprints are distributed with the results. All reported profits use actual costs after full reported uplift.

A. Computational Performance of Parametric CHP

Figure 1 and Table I evaluate the investment in reusable price information. For 30-G1, 1,000 direct queries require 286.46 s, compared with 42.52 s for the complete parametric workflow. The first amortized query is 138. The simpler 30-G2 direction produces eight selected-price segments and reaches amortization at query 18, with a cumulative ratio of 20.54 at 1,000 queries. These differences show that the offer direction and its induced partition matter in addition to system dimensions.

The 118-bus 500-query workload takes 623.41 s by persistent reoptimization and 311.60 s parametrically, reaching

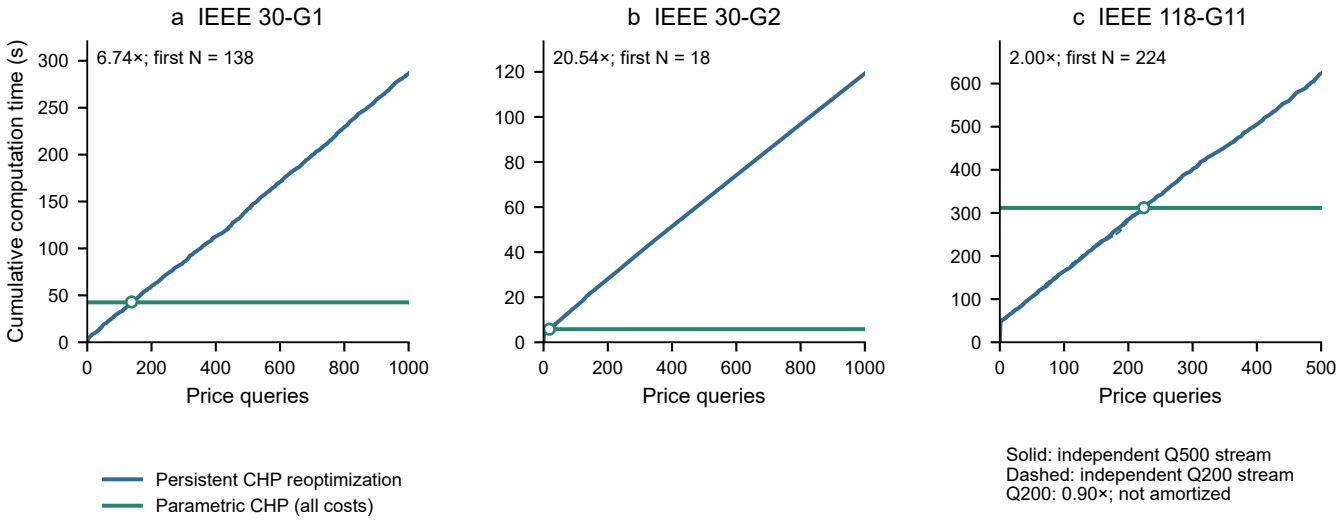


Fig. 1. Measured cumulative CHP pricing time. The two 30-bus streams contain 1,000 queries each. The 118-bus panel distinguishes independent 500- and 200-query workloads. Parametric totals include construction, postprocessing, loading, and measured lookup. Markers identify first observed amortized query counts; the streams are not concatenated.

first measured amortization at query 224. Its independent 200-query comparison takes 279.35 s directly and 311.60 s parametrically. The short workload illustrates the initial mapping investment; the longer workload demonstrates its reuse. Neither its first 200 observations nor its cumulative sequence substitutes for the independent shorter experiment.

The representation also provides structure for strategy evaluation. G1's 154 saved pricing points support 83 compressed selected-price segments. Seventeen common physical regions yield 99 joint intervals, requiring 116 endpoint/physical-plan evaluations at 100 different offers. Proposition 2 converts these into an assessment of the complete declared offer range. The map therefore supplies both repeated pricing and the interval structure needed for economic evaluation.

B. Impact of Strategic Offers on CHP Prices

The profitable all-day offer works through a nonuniform hourly price response. Figure 2(a) shows a marked decrease in hour 1 alongside increasing prices in other hours. Integrating the price effect in (39) gives $+8{,}741.60$ dollars. Hours 2, 4, 5, and 23 each contribute approximately $1{,}232.38$, whereas hour 1 contributes $-2{,}053.58$ and the peak-demand hour contributes about -210.51. The relevant economic weight is the unit's best self-scheduled generation at each price, not demand level alone. Temporal price influence becomes a net gain only after these positive and negative hourly effects are combined.

Bus 1 is the PTDF reference and the location of G1. Consequently, its explicit congestion component is zero, and the original-reference accounting assigns the $8{,}741.60$ price effect to the system energy component. This balance multiplier is formed in the constrained network optimization. Re-expressing the same nodal prices with bus 23 as reference allocates $8{,}530.16$ to energy and 211.44 to congestion, with the same total. Nodal price differences provide a reference-invariant view of the network's price effect.

TABLE II
SETTLEMENT CHANGES AT G1'S BEST OFFER (USD)

Unit	ΔR_g	ΔU_g	ΔC_g	$\Delta\Pi_g$
G1	4385.61	−578.36	−4004.31	7811.56
G2	202.68	−0.10	252.45	−49.87
G3	−5592.77	−508.42	−6742.11	640.92
G4	0.00	0.00	0.00	0.00
G5	11707.21	−312.43	11394.79	0.00
G6	13.64	−13.64	0.00	0.00

Best offer minus truthful bidding under the same higher-profit boundary schedule. Each row satisfies $\Delta\Pi_g = \Delta R_g + \Delta U_g - \Delta C_g$. Other units keep their offers fixed.

Figure 2(b) provides this view at G5, located at bus 23. Near the critical offer in hour 19, the energy component is 114.8846 USD/MWh and the congestion component is -58.5446, yielding a nodal CHP of 56.34. Lines 29 and 30 are the pricing-active lines along the studied path. Their reverse-capacity multipliers account for approximately -14.62 and -43.92 USD/MWh of G5's congestion component in this hour. This identifies marginal network restrictions in the pricing model separately from the physical dispatch.

The selected prices are continuous across the region boundaries, so the complete profit path in (43) has continuous price effects and separately evaluated physical-execution jumps. The energy and congestion components recover the original nodal prices and the integrated price effect.

C. Strategic Profits and Uplift Payments

G1 earns additional profit while its uplift payment falls. Figure 3 evaluates the full offer range and identifies the highest

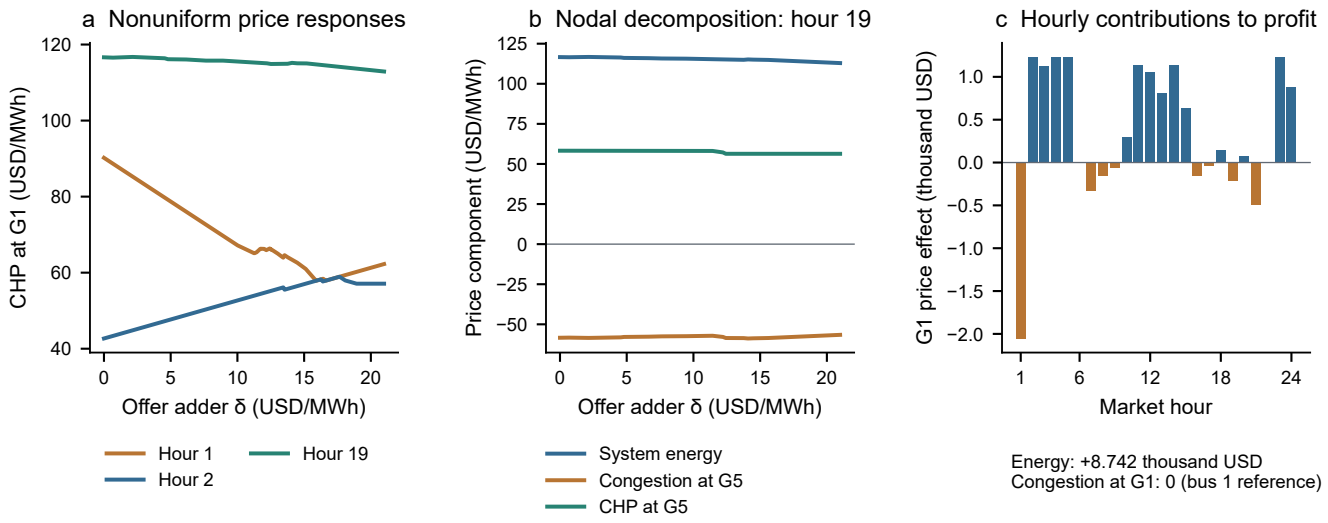


Fig. 2. Strategic-offer price responses. (a) G1-node CHP in hours 1, 2, and 19. (b) System energy, congestion, and total CHP at the G5 node in hour 19. (c) Signed hourly contributions to G1's price-related profit effect along the path to the best offer. The PTDF reference is bus 1, where G1 is located. Hour 2 is one of the tied largest positive contribution hours; hour 19 is the baseline demand peak.

candidate at $\delta^{\Pi} = 13.2040804436$ USD/MWh, corresponding to a 31.4518% normalized adder. Under the upper adjacent-plan rule, actual profit rises from 37,289.31 to 45,100.87: an increment of 7,811.56, or 20.95% of baseline profit. Uplift falls from 594.61 to 16.26. A directly evaluated interior offer $\delta^{\mathrm{UC}} - 0.001$ earns 7,810.84 additional profit on the incoming physical branch. Thus the high-profit opportunity extends to a nearby offer and does not require the producer to control the ISO's choice at the exact cost tie.

Table II explains the accounting. G1 energy revenue increases by 4,385.61, uplift decreases by 578.36, and actual cost decreases by 4,004.31. Their combination produces the observed profit increment. Across the full path, the alternative mechanism decomposition is

$$\Delta\Pi_{G1} = 8741.60 - 29.23 - 900.82 = 7811.56. \tag{51}$$

The terms are the price effect, the integrated offer-exposure effect, and two physical switching effects. Price gains accumulate as the offer increases, while replacement of offer-exposed output produces discrete losses. This is the mechanism by which a favorable side of a physical boundary becomes the most profitable candidate. The plotted branches use common optimal self-schedules over all 99 joint intervals and agree with the finite candidate settlements.

The other producers' accounts show why revenue changes alone do not describe the distribution of gains. G3's actual cost falls more than its revenue, so it earns more profit. G5 sells more energy, but its additional cost and lower uplift offset the revenue increase, leaving its net profit essentially unchanged. Their own offers remain fixed. At system level, total uplift falls from 2,814.52 to 1,401.57, and the transmission adjustment falls from 548.26 to 334.55. The network settlement identity (50) reconciles these payments with the reported UC–CHP cost gap. Reduced total compensation can therefore coexist with a profitable unilateral offer.

The temporal extent of the strategy changes the opportunity available to the same producer. Raising only G1's hour-1 incremental offer over the same normalized 0–50% range has a maximum additional profit of zero under its selected-price and physical-execution rules. Two common physical regions and four endpoint/plan candidates cover that range. At the largest adder, actual profit is approximately 106.42 below baseline even though uplift rises to 732.67. Compared with the profitable all-day direction, this result connects temporal vulnerability to the timing of strategic decisions: a producer's opportunity depends on which hourly offers it can change and the resulting full-horizon settlement.

D. Physical and Network Drivers of Market Power

The critical offer reflects the cost of replacing G1 generation. In schedule A, G5 first operates in hour 6; in schedule B, it operates from hour 1. The additional five online hours provide 125 MWh and replace 105.069565 MWh from G1 and the remainder from G3. Both schedules start G5 once. Their actual system costs are 184,189.786589 and 185,577.133581, and their G1 offer-exposed quantities are 1464.060870 and 1358.991304 MWh. Equation (44) therefore predicts

$$\delta^{\mathrm{UC}} = \frac{1387.346991}{105.069565} = 13.204080444 \text{ USD/MWh}. \tag{52}$$

This physical threshold coincides with the highest upper-branch profit candidate in this case: $\delta^{\Pi} = \delta^{\mathrm{UC}}$. At the common offer and selected CHP, the 1,387.35 actual-profit difference between executions is exactly the additional actual system cost of replacing G1, as predicted by (38). Both executions minimize the ISO's reported objective, yet they allocate different actual profits to G1. Across all 17 archived common physical

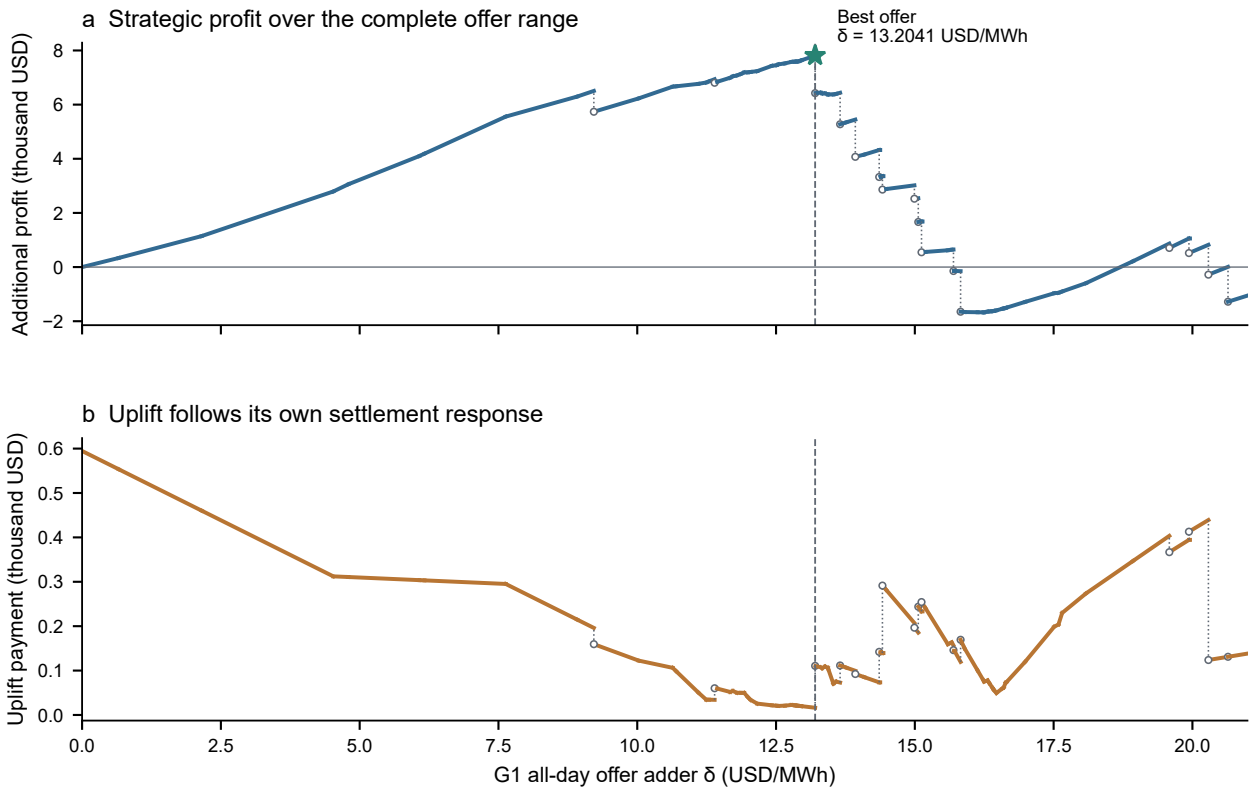


Fig. 3. G1 additional actual profit and uplift payment over the complete offer domain. Continuous pieces use common optimal physical and self-schedules. Dotted vertical indicators and open markers distinguish adjacent limits at physical switches; they do not interpolate attainable intermediate payoffs. The star denotes the declared higher-profit boundary candidate.

regions, the offer-exposed quantity is nonincreasing. All 16 adjacent boundary contrasts satisfy (42); their reported system-cost tie residuals are below 4×10^{-11} dollars. This verifies the sign and magnitude of the pure physical losses separately from the price gains in Fig. 3.

Figure 4(c) tests the cost mechanism. Increasing G5's actual online cost by ϵ dollars per online hour gives

$$\delta^{\mathrm{UC}}(\epsilon) = 13.204080444 + 0.047587519656\epsilon. \tag{53}$$

The predefined $\epsilon = -1, 0, 1, 5$ cases yield thresholds 13.15649, 13.20408, 13.25167, and 13.44202 USD/MWh. Global UC bounds at each crossing and neighboring offers support the candidate schedules. Increasing actual startup cost by \$50 per start leaves the threshold unchanged, as predicted by equal startup counts. These comparisons identify extra online operation as the relevant substitution expense in this transition.

The line-30 intervention compares the offer's effect in two feasible networks. At the fixed offer $\delta^{\mathrm{UC}}-0.001$, relaxing line 30 lowers the hour-19 energy component from 114.8846 to 104.2192 USD/MWh. G5's congestion component rises from -58.5446 to -45.9592, so its nodal CHP increases from 56.34 to 58.26 despite the lower energy component. G5's cleared hour-19 output increases from 41.42 to 73.01 MW. The line-30 pricing multiplier becomes zero, while another line restriction remains active.

The response to the offer differs from this comparison of price levels. In hour 19, G1's price changes by -1.7543 USD/MWh from truthful bidding in the original network but by $+1.9113$ after relaxation: $\mathcal{I}^{\lambda}_{G1,19} = 3.6656$ USD/MWh. G5's price response remains -1.92 in both networks. In hours 2, 4, and 5, G1's response remains $+13.2031$ in both. The network intervention therefore preserves some major positive hourly responses while reversing another response's sign; it does not uniformly attenuate the offer's price influence.

The relaxed system can also use a substitute schedule with a line-30 flow magnitude of approximately 39.01 MW, exceeding the original 20.8 MW limit. The original A/B schedules remain feasible and retain their mutual cost intersection, but the new schedule dominates them by approximately 2,196.5 near that intersection. Transmission expansion changes the set of economic competitors that constrain G1's offer.

Finally, use each network's own truthful baseline. G1 baseline profit is 37,289.31 in the original network and 31,911.46 after relaxation. At $\delta^{\mathrm{UC}} - 0.001$, total profit falls from 45,100.15 to 40,302.15, while additional profit rises from 7,810.84 to 8,390.69. Equation (47) gives the cross-network change

$$\mathcal{I}^{\Pi}_{G1} = -2127.85 + 59.92 + 2647.77 = 579.84 \text{ USD}. \tag{54}$$

Relative to the respective baselines, G1 output falls by 95.00 MWh in the original network and 157.50 MWh after re-

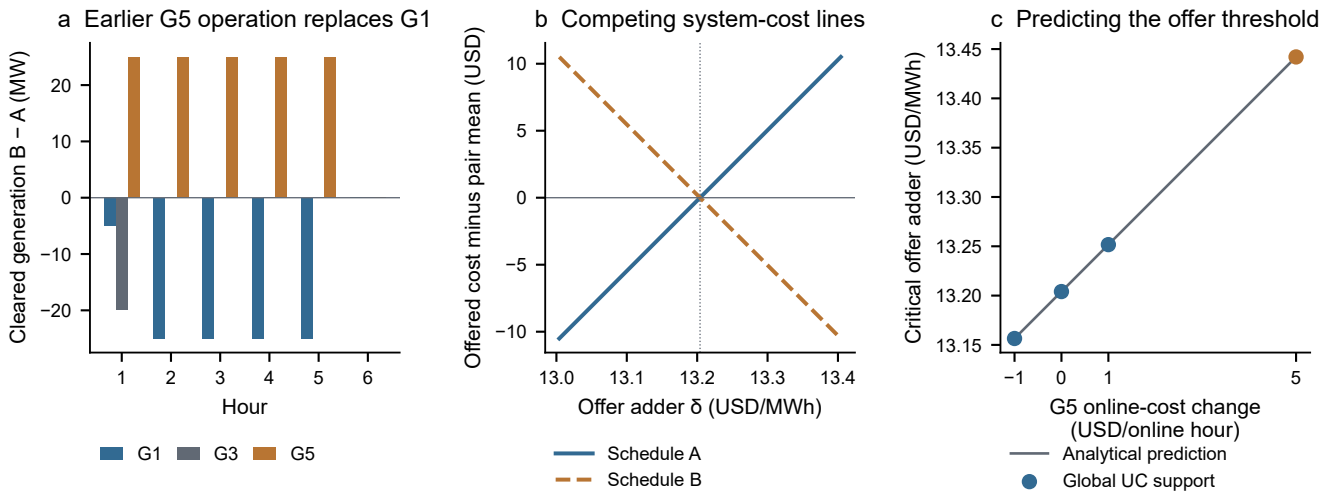


Fig. 4. Physical substitution and the UC switching threshold δ^{UC}. (a) Cleared generation changes from schedule A to B in the first six hours. (b) Competing offered system-cost lines after subtracting their common offer-dependent pair mean, which preserves their crossing. (c) Predicted physical thresholds under specified changes in G5's actual online cost, supported by global UC checks.

laxation. Greater generation replacement thus coexists with higher additional profit at this offer: the larger actual-cost saving exceeds the reduction in incremental energy revenue. This comparison changes the entire pricing and execution problem, whereas Corollary 1 holds within a fixed feasible set. Immediately above the original threshold, additional profits are 6,424.82 and 8,392.45. These are two fixed-offer comparisons, not a calculation of the relaxed network's maximum over the full offer domain.

VI. Conclusion

Continuous strategic offers can be evaluated by jointly representing selected CHP prices and executable UC schedules over the offer range. The network-flow pricing model supplies the common LP structure, while physical cost competition identifies the schedules used for settlement. Their joint intervals support finite profit candidates and reusable price queries, with preprocessing costs accounted for in the measured workloads. Boundary maxima and approachable one-sided profits are distinct outputs of the same evaluation.

The market interpretation follows from combining prices with reported uplift and actual operating cost. At continuous selected prices, increasing a nonnegative offer cannot create a positive profit jump through physical switching alone. When reported system costs tie, the producer's execution-dependent profit difference equals the actual system-cost difference with reversed sign. Replacement costs predict the tested switching threshold and its response to online-cost changes. Across networks, greater replacement need not imply smaller additional profit: the fixed-offer line relaxation increases that gain by 579.84 dollars because actual-cost savings outweigh reduced incremental revenue. These results turn price and dispatch responses into explicit settlement and substitution tests within the specified strategy set.

Data and Code Availability

The result tables, plotting data, implementation, numerical verification records, and supplementary self-scheduling witnesses are available in the CHP_pricing project repository, under `chp_marketpower/paper_results`. The accompanying manuscript record identifies the source versions and the independent query workloads.

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